# Hahn–Banach Norm-Preserving Extension

# An End-to-End Meta-Math Walkthrough

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Abstract

The key finite step extends a dominated functional by one direction. A chain-union argument then extends it to the entire space without claiming countably many directions suffice. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

**FA-05 / Functional Analysis and Measure Theory / Report v1**

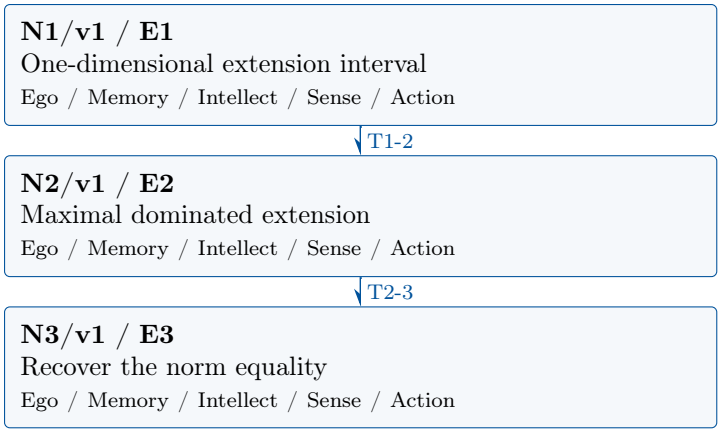
## I. TARGET AND IMPORT CONTRACT

**Theorem I.1.** *Let $X$ be a real normed space, $M \subset X$ any linear subspace, and $f : M \to \mathbb{R}$ bounded linear. There exists bounded linear $F : X \to \mathbb{R}$ with $F|_M = f$ and $\|F\| = \|f\|$. Neither completeness of $X$ nor closedness of $M$ is assumed.*

### A. Foundations and permitted imports

Work in set theory with Zorn's lemma: a nonempty partially ordered set in which every chain has an upper bound has a maximal element. Let $C = \|f\|$ and $p(x) = C\|x\|$, a finite-valued sublinear function: $p(x + y) \leq p(x) + p(y)$ and $p(tx) = tp(x)$ for $t \geq 0$. Use completeness of the real order to select a number between a bounded supremum and infimum.

## II. NODE INTERACTION WORKFLOW




Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

### A. Versioned evidence-transfer ledger

**T1-2: N1/E1/v1 to N2.** $g$ extends linearly to $D + \mathbb{R}z$ while remaining dominated by $p$.

**Receiving activity.** N2.2

**Compatibility.** E1/v1 and all extensions of $f$ dominated by $p$. Check: A union is a function only because the extensions agree on overlaps; a maximal proper domain can still be extended by E1.

**T2-3: N2/E2/v1 to N3.** There is a linear $F : X \to \mathbb{R}$ extending $f$ with $F \leq p$.

**Receiving activity.** N3.1

**Compatibility.** E2/v1 and $p(x) = C\|x\|$. Check: One-sided domination alone must be converted to an absolute bound; the zero-functional case is included.

## III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

### A. N1/v1: One-dimensional extension interval

**Object and role.** Turn domination of $g(d) + ta$ into compatible lower and upper bounds for the new value $a = G(z)$. **Local obligation:** $g$ extends linearly to $D + \mathbb{R}z$ while remaining dominated by $p$.

**Inputs.** A linear $g : D \to \mathbb{R}$ with $g \le p$ on a subspace $D$, and $z \notin D$.

**E — Ego**

Publish E1/v1 only after establishing this obligation: $g$ extends linearly to $D+\mathbb{R}z$ while remaining dominated by $p$.

**M — Memory**

A linear $g : D \to \mathbb{R}$ with $g \le p$ on a subspace $D$, and $z \notin D$. Retain the stated assumptions and import versions.

**I — Intellect**

Turn domination of $g(d)+ta$ into compatible lower and upper bounds for the new value $a = G(z)$.

**S — Sense**

Both signs of $t$ must be checked, and the choice interval must be nonempty and finite.

**A — Action**

Carry out activities N1.1–N1.3 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N1.1 For $u, v \in D$, linearity and domination give $g(u) + g(v) = g(u + v) \le p(u + v) \le p(u - z) + p(v + z)$. Therefore $g(u)-p(u-z) \le p(v+z)-g(v)$. Set $L = \sup_u[g(u)-p(u-z)]$ and $U = \inf_v[p(v+z)-g(v)]$. The preceding inequality gives $L \le U$; choosing $u = 0$ and $v = 0$ bounds both endpoints between finite numbers.

N1.2 Choose $a \in [L, U]$ and define $G(d + tz) = g(d) + ta$. The representation is unique because $z \notin D$, so $G$ is well defined and linear. For $t > 0$, $a \le p(d/t + z) - g(d/t)$ implies $G(d + tz) \le p(d + tz)$.

N1.3 For $t = -s < 0$, $a \ge g(d/s) - p(d/s - z)$ gives $g(d) - sa \le sp(d/s - z) = p(d - sz)$. The case $t = 0$ uses $g \le p$. Thus domination holds for every real scalar.

**Export E1/v1.** $g$ extends linearly to $D + \mathbb{R}z$ while remaining dominated by $p$.

**Acceptance gate.** Both signs of $t$ must be checked, and the choice interval must be nonempty and finite.

**Targeted repair.** If only positive scalars were considered, restore the lower endpoint $L$ and the negative-scalar calculation before extending the domain. Withhold this export and recheck its descendants after a versioned correction.

## B. N2/v1: Maximal dominated extension

**Object and role.** Order compatible extensions by domain inclusion and take unions along chains.
**Local obligation:** There is a linear $F : X \to \mathbb{R}$ extending $f$ with $F \leq p$.

**Inputs.** E1/v1 and all extensions of $f$ dominated by $p$. Incoming proof exports: E1/v1 at N2.2.

**E — Ego**

Publish E2/v1 only after establishing this obligation: There is a linear $F : X \to \mathbb{R}$ extending $f$ with $F \leq p$.

**M — Memory**

E1/v1 and all extensions of $f$ dominated by $p$. Incoming proof exports: E1/v1 at N2.2. Retain the stated assumptions and import versions.

**I — Intellect**

Order compatible extensions by domain inclusion and take unions along chains.

**S — Sense**

A union is a function only because the extensions agree on overlaps; a maximal proper domain can still be extended by E1.

**A — Action**

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N2.1 The set of pairs $(D, g)$ with $M \subset D \subset X$, $g|_M = f$, and $g \leq p$ is nonempty. Order it by extension. For a chain, the union of its domains is a subspace: any two vectors and their values occur together in one chain member. The union map is consequently well defined, linear, and dominated by $p$.

N2.2 This union is an upper bound, so Zorn supplies a maximal pair $(D, g)$. If $D \neq X$, choose $z \in X \setminus D$ and apply E1, contradicting maximality. Thus $D = X$; denote the resulting map by $F$.

**Export E2/v1.** There is a linear $F : X \to \mathbb{R}$ extending $f$ with $F \leq p$.

**Acceptance gate.** A union is a function only because the extensions agree on overlaps; a maximal proper domain can still be extended by E1.

**Targeted repair.** If a union of incomparable extensions is used, restrict to chains before invoking Zorn; arbitrary extensions can assign conflicting values. Withhold this export and recheck its descendants after a versioned correction.

### C. N3/v1: Recover the norm equality

**Object and role.** Apply the one-sided domination to $x$ and $-x$, then compare with the restriction.
**Local obligation:** $F|_M = f$ and $\|F\| = \|f\|$.

**Inputs.** E2/v1 and $p(x) = C\|x\|$. Incoming proof exports: E2/v1 at N3.1.

**E — Ego**

Publish E3/v1 only after establishing this obligation: $F|_M = f$ and $\|F\| = \|f\|$.

**M — Memory**

E2/v1 and $p(x) = C\|x\|$. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

**I — Intellect**

Apply the one-sided domination to $x$ and $-x$, then compare with the restriction.

**S — Sense**

One-sided domination alone must be converted to an absolute bound; the zero-functional case is included.

**A — Action**

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N3.1 For every $x$, $F(x) \le C\|x\|$ and $-F(x) = F(-x) \le C\|x\|$. Hence $|F(x)| \le C\|x\|$, so $F$ is bounded and $\|F\| \le C$.

N3.2 Restricting $F$ to the unit ball of $M$ gives $\|f\| \le \|F\|$. Since $C = \|f\|$, equality follows. If $C = 0$, this argument yields $F = 0$ and still applies.

**Export E3/v1.** $F|_M = f$ and $\|F\| = \|f\|$.

**Acceptance gate.** One-sided domination alone must be converted to an absolute bound; the zero-functional case is included.

**Targeted repair.** If the norm conclusion is inferred directly from $F \le p$, add the $-x$ bound and the restriction inequality. Withhold this export and recheck its descendants after a versioned correction.

## IV. CONTINUOUS MATHEMATICAL PROOF

### A. N1 — One-dimensional extension interval

1. **N1.1.** For $u, v \in D$, linearity and domination give $g(u)+g(v) = g(u+v) \le p(u+v) \le p(u-z)+p(v+z)$. Therefore $g(u)-p(u-z) \le p(v+z)-g(v)$. Set $L = \sup_u[g(u)-p(u-z)]$ and $U = \inf_v[p(v+z)-g(v)]$. The preceding inequality gives $L \le U$; choosing $u = 0$ and $v = 0$ bounds both endpoints between finite numbers.

2. **N1.2.** Choose $a \in [L, U]$ and define $G(d+tz) = g(d)+ta$. The representation is unique because $z \notin D$, so $G$ is well defined and linear. For $t > 0$, $a \le p(d/t+z)-g(d/t)$ implies $G(d+tz) \le p(d+tz)$.

3. **N1.3.** For $t = -s < 0$, $a \ge g(d/s)-p(d/s-z)$ gives $g(d)-sa \le sp(d/s-z) = p(d-sz)$. The case $t = 0$ uses $g \le p$. Thus domination holds for every real scalar.

### B. N2 — Maximal dominated extension

1. **N2.1.** The set of pairs $(D, g)$ with $M \subset D \subset X$, $g|_M = f$, and $g \le p$ is nonempty. Order it by extension. For a chain, the union of its domains is a subspace: any two vectors and their values occur together in one chain member. The union map is consequently well defined, linear, and dominated by $p$.

2. **N2.2.** This union is an upper bound, so Zorn supplies a maximal pair $(D, g)$. If $D \ne X$, choose $z \in X \setminus D$ and apply E1, contradicting maximality. Thus $D = X$; denote the resulting map by $F$.

### C. N3 — Recover the norm equality

1. **N3.1.** For every $x$, $F(x) \le C\|x\|$ and $-F(x) = F(-x) \le C\|x\|$. Hence $|F(x)| \le C\|x\|$, so $F$ is bounded and $\|F\| \le C$.

2. **N3.2.** Restricting $F$ to the unit ball of $M$ gives $\|f\| \le \|F\|$. Since $C = \|f\|$, equality follows. If $C = 0$, this argument yields $F = 0$ and still applies.

□

## V. WORKED ORCHESTRATION

In $\mathbb{R}^2$ with the Euclidean norm, take $M = \{(t, t)\}$ and $f(t, t) = 2t$. The norm on $M$ is $\sqrt{2}$. The extension $F(x, y) = x + y$ has norm $\sqrt{2}$ and restricts correctly. This is a concrete certificate; the general construction does not promise a unique extension.

### A. Boundary and hypothesis audit

Uniqueness is false. In $\mathbb{R}^2$ with the $\ell^1$ norm, let $M$ be the first coordinate axis and $f(x,0) = x$. Every $F_a(x,y) = x + ay$ with $|a| \le 1$ extends $f$ and has norm one.

## VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

### A. Rejected argument, diagnosis, and correction

**Rejected N1 candidate:** choose any $a$ below the upper bound and ignore the lower bound. For $X = \mathbb{R}$, $D = \{0\}$, $p(t) = |t|$, choosing $a = -2$ satisfies $a \le 1$ but fails domination at $-1$, where $G(-1) = 2 > 1$. **Repair:** require $a \in [-1, 1]$, the actual two-sided interval. This corrects the extension argument under the same theorem. Recheck the maximal-extension and norm exports.

### B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Deduce that every nonzero vector has a norming continuous linear functional by first defining it on its one-dimensional span.

**Provenance.** This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

**Further reading.** MIT OpenCourseWare, Introduction to Functional Analysis: lecture notes. See the lectures on integration, Hilbert spaces, Baire category, and Hahn–Banach.